Ordinal Depth Estimation for Recursive Language Model Generations: A Depth Degradation Atlas Across Scales and Strategies

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Abstract

Web-scraped training corpora increasingly contain text produced by repeated model-to-data-to-model feedback loops, yet existing detectors only answer whether text is machine-generated, not how many recursive cycles it has undergone. We formalize *ordinal generation depth estimation* as a new task: given a passage, predict its recursive generation depth $d \in \{0, 1, 2, 3\}$. We construct a multi-generational benchmark spanning three model families (1B–14B parameters), three decoding strategies, and four depths, then build the first *Depth Degradation Atlas*, a systematic empirical map of how text statistical properties evolve across recursive depths, models, and strategies. Random Forest classifiers on 19 statistical features extracted from a fixed reference model achieve 70–85% four-class accuracy (chance = 25%). The Atlas reveals that perplexity tail statistics (`p90_ppl`, `var_surprisal`) are universal depth markers across all settings, that inter-depth discriminability decays monotonically (formalized as the Effective Discriminable Depth, EDD), and that temperature sampling induces a feature trajectory reversal that breaks naive cross-strategy transfer. We address this reversal with a two-stage pipeline (strategy identification followed by strategy-conditioned depth classification) that achieves 75.2% mean accuracy, outperforming naive pooling by 3.2 percentage points. The depth signal persists under topic-seeded prompted generation (92.8% accuracy retention) and at the 14B scale without quantization, confirming ecological relevance and scale robustness.

1 Introduction

Large Language Models (LLMs) are now prolific generators of web content. When this content is scraped and fed back into training corpora, it creates recursive feedback loops in which models train on their own outputs, or outputs of earlier model generations. Recent theoretical and empirical work has shown that such loops drive *model collapse*: progressive loss of distributional diversity, tail truncation, and convergence toward low-entropy attractors [Shumailov et al., 2024, Alemohammad et al., 2024, Dohmatob et al., 2024, Gerstgrasser et al., 2024]. The practical consequence is clear. Training data quality degrades as recursive synthetic content accumulates, and curators need tools to detect and manage this contamination.

Existing approaches to machine-generated text detection frame the problem as a binary classification task: is a given passage human-written or machine-generated [Mitchell et al., 2023, Hans et al., 2024, Bao et al., 2024]? While useful, binary labels discard critical information. First-generation synthetic data may be high quality and even beneficial for training [Gunasekar et al., 2023], whereas deeply recursive text is demonstrably degraded. A curator who can distinguish depth-1 from depth-3 text can make graduated decisions (retain, downweight, or discard) rather than applying a single threshold.

No prior method estimates ordinal generation depth from text alone. Model collapse studies characterize degradation patterns but assume depth is known a priori [Drayson et al., 2025, Shumailov et al., 2024]. EditLens [Liang et al., 2024a] estimates single-pass AI editing extent but does not address recursive depth. Watermarking schemes [Kirchenbauer et al., 2023] require generator cooperation and do not survive retraining cycles. There is a methodological gap between knowing that recursive collapse happens and being able to measure how deep a given text sits in the feedback chain.

We address this gap with three contributions. First, we formalize ordinal generation depth estimation as a classification task and construct a controlled multi-generational benchmark spanning three model families (Pythia-1.4B, OLMo-1B, GPT-2 XL), three decoding strategies (nucleus, temperature, top-k), and four recursive depths ($d \in \{0, 1, 2, 3\}$). We show that Random Forest classifiers on 19 statistical features from a fixed reference model achieve 70–85% four-class accuracy across all nine model-strategy configurations. Second, we provide the first *Depth Degradation Atlas*: a systematic empirical characterization of how statistical properties evolve across depths, models, and strategies. The Atlas reveals universal depth markers (perplexity tail statistics consistently rank among the top four features), monotonically decaying inter-depth discriminability (formalized as the Effective Discriminable Depth, EDD), and a strategy-dependent feature trajectory reversal caused by temperature sampling's accelerated convergence to the attractor distribution. We extend this Atlas to the 14B scale (Qwen2.5-14B, clean FP16 without quantization confounds) and to prompted generation (topic-seeded text), confirming that the depth signal is neither scale-limited nor an artifact of unconditional generation. Third, we propose a two-stage pipeline for cross-strategy deployment: a strategy identification classifier (81.2% accuracy on 28 features) routes samples to strategy-conditioned depth classifiers, achieving 75.2% mean accuracy and outperforming naive pooling by 3.2 percentage points.

2 Related Work

Model Collapse and Recursive Generation. Shumailov et al. [2024] demonstrated that iteratively retraining language models on their own outputs leads to progressive distributional collapse and loss of tail diversity. Subsequent work formalized convergence rates [Dohmatob et al., 2024, Gerstgrasser et al., 2024] and showed that even partial synthetic contamination accelerates collapse [Alemohammad et al., 2024]. Drayson et al. [2025] provided empirical characterizations of degradation trajectories across model families. These studies assume generation depth is known, as they control the recursive pipeline. Our work complements this line by estimating depth from text when the generation history is unavailable.

Machine-Generated Text Detection. Binary detectors distinguish human-written from machine-generated text using statistical divergence (DetectGPT [Mitchell et al., 2023], Fast-DetectGPT [Bao et al., 2024]), likelihood ratios (Binoculars [Hans et al., 2024]), or learned classifiers (LOKI [Li et al., 2024a]). DivEye [Liang et al., 2024b] detects synthetic text via n-gram diversity features. These methods produce a single binary or continuous score and do not capture the ordinal structure of recursive generation. Our feature extraction pipeline builds on similar statistical signals (perplexity, surprisal) but targets a multi-class ordinal prediction.

Decoding Strategy Effects. Holtzman et al. [2020] showed that decoding strategy fundamentally shapes the statistical properties of generated text, with nucleus sampling producing more human-like distributions than pure top-k or temperature scaling. Garcés Arias et al. [2025] recently confirmed that strategy effects interact with model architecture in detection tasks. Our Atlas extends this finding to the recursive setting, revealing that temperature sampling accelerates convergence to the attractor distribution, causing feature trajectory reversal relative to nucleus and top-k strategies.

Authorship and Source Attribution. Multi-class attribution of text to specific models or human authors has been studied in stylometry [Uchendu et al., 2020] and multi-source detection [Li et al., 2024b]. These methods identify *which* model generated a text, not *how many recursive cycles* it has undergone.

Data Curation and Quality Scoring. Data selection methods like DSIR [Xie et al., 2023] and QuRating [Wettig et al., 2024] score training examples by distributional proximity to a target or by quality dimensions. Our depth estimates provide a complementary signal: a graduated quality proxy that captures recursive contamination rather than topical fit.

Watermarking. Statistical watermarks [Kirchenbauer et al., 2023] embed detectable signals into generated text. While effective for single-generation attribution, watermarks do not survive the retraining step in recursive feedback loops and require generator cooperation.

3 Task Formalization and Method

3.1 Problem Definition

Let M_0 be a pretrained language model and $\mathcal{D}_0$ a corpus of human-written text. Define the recursive generation chain:

$$\mathcal{D}_{d+1} = \text{Generate}(M_d, S), \quad M_{d+1} = \text{FineTune}(M_0, \mathcal{D}_{d+1}) \tag{1}$$

where S is a decoding strategy (nucleus, temperature, or top-k) and $d \in \{0, 1, 2, \ldots\}$ is the generation depth. Depth 0 denotes human-written text. Given a text passage x with unknown depth d^*, the task is to predict $\hat{d} = f(x)$ where f maps text to an ordinal depth class.

3.2 Benchmark Construction

We construct recursive generation chains for three model families: Pythia-1.4B [Biderman et al., 2023], OLMo-1B [Groeneveld et al., 2024], and GPT-2 XL [Radford et al., 2019]. Each model is fine-tuned using LoRA [Hu et al., 2022] on its own generated text, then generates the next depth's corpus. We apply three decoding strategies: nucleus sampling ($p = 0.95$), temperature sampling ($\tau = 0.9$), and top-k sampling ($k = 50$). This yields a 3×3 grid of 9 model-strategy configurations, each producing depths $d \in \{0, 1, 2, 3\}$. We additionally extend the chain to $d \in \{0, \ldots, 5\}$ for depth limit analysis.

For scale validation, we construct a separate chain using Qwen2.5-14B [Yang et al., 2024] in clean FP16 (no quantization) with LoRA fine-tuning and nucleus sampling. For ecological validity, we generate topic-seeded text using GPT-2 XL conditioned on 50 diverse topic prompts spanning science, politics, creative writing, business, and health domains.

3.3 Feature Extraction

For each passage x, we compute token-level log-probabilities under a fixed reference model (Pythia-1.4B) and extract 19 statistical features organized into four families:

- **Perplexity distribution** (7 features): mean, standard deviation, median, p75, p90, interquartile range, and max of per-token perplexity.
- **Surprisal statistics** (5 features): mean, variance, skewness, kurtosis, and range of per-token surprisal ($-\log p(x_t|x_{<t})$).
- **Diversity metrics** (4 features): type-token ratio, hapax legomena ratio, bigram entropy, and trigram entropy.
- **Fluency indicators** (3 features): mean log-probability, fraction of tokens with $p < 0.01$, and consecutive low-probability token count.

The reference model is held fixed across all experiments to ensure that feature values reflect properties of the text rather than the generating model. When evaluating text generated by Pythia-1.4B itself, we note that cross-scale analysis (Section 5.2) demonstrates that Pythia-1.4B retains discriminative power even on 14B-generated text, with accuracy within 2–4 percentage points of the within-scale setting.

For the two-stage pipeline (Section 5.4), we augment these 19 features with 9 additional strategy-discriminative features: vocabulary coverage ratio, token probability entropy, mean top-1 probability, bigram repetition ratio, hapax legomena ratio (on probability-weighted tokens), top-10 probability mass, Zipf's alpha, unique bigram ratio, and mean token rank.

3.4 Classification

We train Random Forest (RF) classifiers with 500 trees and Logistic Regression (LR) baselines using 5-fold stratified cross-validation. For ordinal structure, we additionally evaluate pairwise binary classifiers at each adjacent depth boundary (d vs $d{+}1$), reporting Area Under the ROC Curve (AUC) as the primary discrimination metric.

3.5 Effective Discriminable Depth (EDD)

We define the Effective Discriminable Depth as the maximum depth d^* such that all adjacent-depth binary classifiers up to d^* exceed an AUC threshold T:

$$\text{EDD}(T) = \max\{d^* : \text{AUC}(d \text{ vs } d{+}1) \geq T,\ \forall\, d < d^*\} \tag{2}$$

We report EDD at thresholds $T \in \{0.65, 0.70, 0.75, 0.80\}$ to characterize sensitivity. At $T = 0.75$, seven of nine model-strategy configurations achieve EDD ≥ 3, indicating that at least three adjacent-depth boundaries are reliably discriminable.

4 The Depth Degradation Atlas

The Depth Degradation Atlas is a systematic empirical map of how text statistical properties evolve across recursive depths. We organize findings along three axes: feature universality, inter-depth discriminability, and strategy-dependent interaction effects.

4.1 Feature Universality

Across all nine model-strategy configurations in the 1B benchmark, `p90_ppl` (90th percentile of per-token perplexity) and `var_surprisal` (variance of per-token surprisal) consistently rank among the top four features by Random Forest importance. This pattern extends to the 14B scale (Table 2), where these two features retain their top-2 ranking (importances 0.185 and 0.162, respectively), and to prompted generation (Table 3), where they again occupy positions 1 and 2 (importances 0.195 and 0.188). The consistency of tail perplexity statistics as depth markers across model families, parameter counts, and generation modes constitutes a core finding of this work.

4.2 Monotonic Discriminability Decay

Adjacent-depth discrimination decays monotonically with depth. In the 1B benchmark, d_0 vs d_1 AUC ranges from 0.981 to 0.998 across all configurations; d_1 vs d_2 AUC ranges from 0.713 to 0.937; d_2 vs d_3 AUC ranges from 0.623 to 0.883. The 14B experiment confirms this pattern: d_0vd_1 AUC = 0.991, d_1vd_2 AUC = 0.867, d_2vd_3 AUC = 0.793 (Table 2). This decay reflects the progressive convergence of the generated distribution toward a low-entropy attractor: each recursive fine-tuning step compresses the distribution, and the marginal compression diminishes as the distribution approaches the attractor.

For depth extension analysis, d_3 vs d_4 AUC clusters near 0.75 across the three 1B models (Pythia: 0.784, $p = 0.001$; OLMo: 0.765, $p = 0.091$; GPT-2 XL: 0.750, $p = 0.505$), while d_4 vs d_5 AUC uniformly falls below 0.70 (0.681–0.694). We report these with full confidence intervals and p-values rather than claiming uniform statistical significance.

4.3 Strategy-Dependent Feature Trajectory Reversal

The Atlas reveals a qualitative interaction between decoding strategy and depth. Under nucleus ($p = 0.95$) and top-k ($k = 50$) sampling, features evolve monotonically with depth: perplexity tail statistics decrease, diversity metrics contract, and surprisal variance narrows. Temperature sampling ($\tau = 0.9$) preserves the direction at the d_0/d_1 boundary but reverses the d_1/d_2 feature trajectory relative to the other

Table 1: Four-class Random Forest accuracy (%) on the 1B benchmark. All values from 5-fold stratified cross-validation. Chance level = 25%.

Model	**Nucleus**	**Temp (τ=0.9)**	**Top-k (50)**
Pythia-1.4B	78.2	76.8	72.4
OLMo-1B	78.8	74.5	69.6
GPT-2 XL	83.3	85.4	80.1

strategies. This reversal is visible in cross-strategy leave-one-out (LOO) experiments: when a classifier trained on nucleus and top-k data is evaluated on temperature-sampled text, the d_1 vs d_2 AUC drops to 0.13 (GPT-2 XL) and 0.25 (OLMo), indicating that the decision boundary is oriented in the wrong direction.

The root cause is a genuine structural distributional shift. Temperature scaling uniformly divides logits by τ, flattening the probability distribution across the entire vocabulary. In contrast, nucleus and top-k truncate the distribution, preserving the shape of the included portion. Uniform flattening accelerates convergence to the attractor distribution: temperature depth-N features approximate nucleus/top-k depth-$(N{+}1)$ features. For GPT-2 XL, the median `p90_ppl` at temperature depth 1 (92.2) falls squarely within the nucleus/top-k depth-2 range (65–92).

This shift cannot be resolved by simple feature normalization. Z-score standardization improves overall LR cross-strategy accuracy from 25% to 61%, but does not recover pairwise d_1vd_2 AUC (GPT-2 XL: 0.50 $\rightarrow$ 0.24; OLMo: 0.50 $\rightarrow$ 0.19). The worsening AUC under normalization confirms a geometric rotation of the feature space, not a rescalable offset. Whether more sophisticated domain adaptation methods could recover this boundary remains an open question.

We categorize the three strategies into distinct tiers: nucleus serves as a stable baseline with consistent feature trajectories; temperature enhances signal magnitude at shallow depths but induces the d_1/d_2 reversal in cross-strategy settings; top-k attenuates signal strength but preserves directional consistency.

5 Experiments

5.1 1B Benchmark: 3-Model $\times$ 3-Strategy Grid

Table 1 summarizes four-class RF accuracy across the nine configurations in the 1B benchmark. Accuracy ranges from 69.6% (OLMo, top-k) to 85.4% (GPT-2 XL, temperature), with all configurations substantially above the 25% chance level. GPT-2 XL achieves the highest accuracy across all three strategies, likely due to its larger effective capacity among the 1B-class models.

Per-class recall shows a consistent pattern: d_0 recall exceeds 97% in all settings (the human-vs-machine boundary is easy), while d_2 recall is the lowest (26.8–63.0%), reflecting the compressed inter-depth distances at middle depths. Three-class accuracy excluding d_0 ranges from 59% to 82%, providing a conservative estimate of ordinal depth estimation performance isolated from the binary detection component.

5.2 Scale Validation: 14B Clean FP16

Prior experiments at the 7B scale using QLoRA produced contradictory results (non-monotonic accuracy curves), raising concerns about whether the depth signal is an artifact of small models. We resolve this with a clean experiment: Qwen2.5-14B fine-tuned with LoRA in FP16 precision (no quantization), using nucleus sampling ($p = 0.95$).

Table 2 presents the results. The 14B model achieves 76.2% four-class RF accuracy [95% CI: 74.8–77.6%], falling within the range established by the 1B models (69.6–85.4%). The EDD at $T = 0.75$ is 3, confirming that three adjacent-depth boundaries remain discriminable. Feature importance rankings preserve `p90_ppl` and `var_surprisal` as the top two features (importances 0.185 and 0.162). No

Table 2: Scale comparison: 14B clean FP16 vs 1B models (nucleus sampling, 4-class RF). LLaMA-3 8B IT included as an intermediate reference.

Model	**Scale**	**4-cl Acc**	d_1vd_2	d_2vd_3	**EDD**
Pythia-1.4B	1.4B	78.2%	0.861	0.801	4
OLMo-1B	1B	78.8%	0.867	0.812	4
GPT-2 XL	1.5B	83.3%	0.913	0.883	4
LLaMA-3 8B IT	8B	70.2%	0.877	0.771	4
Qwen2.5-14B	14B	76.2%	0.867	0.793	3

Table 3: Prompted vs BOS unconditional generation (GPT-2 XL, nucleus). Retention is the ratio of prompted to BOS accuracy.

Setting	**4-cl Acc**	d_1vd_2	d_2vd_3	**Top-2 Features**
BOS (unconditional)	85.4%	0.937	0.882	p90_ppl, var_surp
Prompted (topic-seeded)	79.2%	0.896	0.852	p90_ppl, var_surp
Retention	92.8%	95.6%	96.6%	identical ranking

U-shaped artifact appears, confirming that the prior 7B anomalies were quantization noise rather than a scale effect.

A notable finding is that Pythia-1.4B, used as the reference model, retains discriminative power on 14B-generated text. The cross-scale accuracy gap is only 2–4 percentage points relative to within-scale performance, indicating that the depth signal manifests as a distributional property of the text itself rather than a model-specific artifact.

5.3 Prompted Generation: Ecological Validity

Real-world machine-generated text is almost always conditioned on some prompt or context. If the depth signal exists only under unconditional (BOS-seeded) generation, its practical value is limited. We test this by generating topic-seeded text with GPT-2 XL using 50 diverse prompts (e.g., "The following article discusses climate change…") across five domain clusters.

Table 3 shows that prompted generation retains 92.8% of the BOS baseline accuracy (79.2% vs 85.4%). Feature rankings are identical: `p90_ppl` and `var_surprisal` occupy positions 1 and 2 with importances of 0.195 and 0.188. Kolmogorov-Smirnov tests confirm that the distributions of the top-5 features do not differ significantly between BOS and prompted generation ($p > 0.05$ for all five features; e.g., `p90_ppl` $p = 0.23$, `var_surprisal` $p = 0.41$).

Table 4 breaks down accuracy by topic cluster. Cross-topic variance is small (standard deviation = 3.1 percentage points), with the worst cluster (Creative/Narrative, 74%) still far above chance. This robustness to content domain is expected: topic conditioning constrains semantic content but does not alter the recursive fine-tuning mechanism that produces depth-related distributional shifts.

5.4 Cross-Strategy Two-Stage Pipeline

Naive cross-strategy transfer suffers from the feature trajectory reversal described in Section 4.3: leave-one-out cross-strategy accuracy drops to 59.1% on average, with temperature holdout showing catastrophic d_1vd_2 inversion. We address this with a two-stage pipeline.

Stage 1: Strategy Identification. Using the original 19 depth-oriented features, strategy classification accuracy is only 57.5% (v1). The depth features were not designed to distinguish decoding strategies. We augment them with 9 strategy-discriminative features (Section 3.3) that capture probability distribution shape: `top1_prob_mean`, `vocab_coverage_ratio`, `prob_mass_top10`,

Table 4: Per-topic cluster accuracy for prompted generation (GPT-2 XL, nucleus, 4-class RF).

Topic Cluster	n **prompts**	**4-cl Acc**	d_1**v**d_2 **AUC**
Science/Technology	12	81%	0.91
Politics/Society	10	78%	0.88
Creative/Narrative	8	74%	0.85
Business/Finance	10	80%	0.90
Health/Lifestyle	10	84%	0.92

Table 5: Strategy classification accuracy: v1 (19 depth features) vs v2 (28 features including strategy-discriminative additions).

Model	**v1 (19 feat)**	**v2 (28 feat)**	**Δ**
Pythia-1.4B	52.3%	78.3%	+26.0
OLMo-1B	54.8%	79.8%	+25.0
GPT-2 XL	65.4%	85.6%	+20.2
Mean	57.5%	81.2%	+23.7

`rep_bigram_ratio`, and others. These features directly encode the distributional differences between strategies: temperature lowers top-1 probability and raises entropy; top-k restricts vocabulary coverage; nucleus produces an intermediate profile.

Table 5 shows that the enhanced feature set (v2, 28 features) improves mean strategy classification from 57.5% to 81.2%, a gain of 23.7 percentage points. GPT-2 XL achieves the highest accuracy (85.6%), consistent with its larger inter-strategy feature gaps.

Stage 2: Strategy-Conditioned Depth Classification. After strategy identification, samples are routed to strategy-matched depth classifiers. Table 6 compares the full pipeline against baselines. The predicted two-stage pipeline (v2) achieves 75.2% mean accuracy, outperforming naive pooled classification (72.0%) by 3.2 percentage points. The oracle two-stage pipeline (perfect strategy identification) reaches 78.4%, nearly identical to the within-strategy upper bound (78.2%), confirming that the pipeline architecture is sound and that remaining error is dominated by strategy misclassification.

The gap between predicted (75.2%) and oracle (78.4%) is 3.2 percentage points. Closing this gap requires improving strategy classification beyond 81.2%, a tractable engineering challenge. Temperature recall (73–79%) is the weakest link, as temperature and nucleus produce partially overlapping probability shapes at intermediate depths.

5.5 Wild-Data Calibration

To bridge controlled benchmarks with real-world deployment, we apply trained RF classifiers to passages from RedPajama-v2 and Dolma, two large-scale web corpora without ground-truth depth labels. Table 7 compares depth distributions across six corpora of varying quality levels.

The results show a consistent gradient: editorially curated text (WikiText-103) is classified as 100% d_0; curated web (OpenWebText) shows 1.4% d_1+; filtered web (Dolma) shows 14.9%; and raw web crawls (RedPajama-v2, C4, FineWeb) show 17–20% d_1+. This gradient aligns with the expected quality hierarchy and with the known presence of template, SEO, and low-diversity content in web crawls, which produces perplexity signatures that overlap with recursive synthetic text.

Cross-model consistency is high: pairwise agreement between classifiers trained on different reference configurations ranges from 87% (Pythia vs GPT-2 XL) to 91% (Pythia vs OLMo). Three-model unanimous classification identifies 78.5% of passages as d_0 and 8.2% as d_1+, providing a conservative lower bound on synthetic contamination.

Table 6: Two-stage pipeline accuracy comparison (4-class depth estimation). Cross-strategy LOO is the naive transfer baseline; within-strategy is the upper bound.

Method	Pythia	OLMo	GPT-2 XL	Mean
Cross-strategy LOO		—		59.1%
Naive pooled	70.9%	71.2%	73.9%	72.0%
Predicted 2-stage (v1)	69.3%	69.5%	73.0%	70.6%
Predicted 2-stage (v2)	**73.7%**	**74.8%**	**77.2%**	**75.2%**
Oracle 2-stage	74.6%	78.5%	82.0%	78.4%
Within-strategy	74.5%	78.4%	81.6%	78.2%

Table 7: Wild-data depth distribution comparison. FPR estimate reflects proportion classified as d_1+ in corpora assumed to be predominantly human-written.

Dataset	Quality	d_0%	d_1+%	FPR est.
WikiText-103	Editorial	100	0	0%
OpenWebText	Curated web	98.6	1.4	$\sim$1%
Dolma	Filtered web	85.1	14.9	$\sim$15%
RedPajama-v2	Web crawl	82.3	17.7	$\sim$18%
C4	Web crawl	79.8	20.2	$\sim$20%
FineWeb	Web crawl	80.5	19.5	$\sim$19%

A temporal analysis reveals a striking pattern: pre-2022 web text is classified as 94.2% d_0, while post-2023 text drops to 76.8% d_0, a 17.4 percentage point shift consistent with the timeline of widespread LLM deployment. While this does not constitute causal evidence, the correlation directly supports the practical motivation for ordinal depth estimation tools.

6 Atlas Visualizations

We present four key visualizations that constitute the empirical core of the Depth Degradation Atlas.

7 Discussion

Why Ordinal, Not Binary? The monotonic decay in adjacent-depth AUC (Figure 1) demonstrates that ordinal depth carries information beyond the binary real-vs-synthetic distinction. While d_0 vs d_1 is the easiest boundary (AUC > 0.99), the d_1 vs d_2 and d_2 vs d_3 boundaries remain discriminable (AUC 0.79–0.94), enabling graduated curation decisions. A binary detector collapses all synthetic depths into one category, discarding the distinction between potentially useful first-generation text and demonstrably degraded deeper-generation text.

Depth-Graduated Curation. Predicted depth provides a finer-grained quality proxy than binary labels. A corpus curator could retain d_0 and d_1 text while downweighting d_2 and discarding d_3+, or apply depth-proportional sample weights during training. The monotonic relationship between depth and distributional degradation supports such graduated strategies. Full experimental validation of depth-weighted training is left to future work.

Cross-Strategy d_1vd_2 AUC. Table 8 reports the per-strategy d_1vd_2 AUC in cross-strategy LOO evaluation. The temperature holdout exhibits AUC values of 0.13 (GPT-2 XL) and 0.25 (OLMo), below 0.5, indicating that the classifier's decision boundary is oriented opposite to the true boundary. This is not noise: the mechanism (temperature's uniform logit scaling accelerating convergence) is identified and

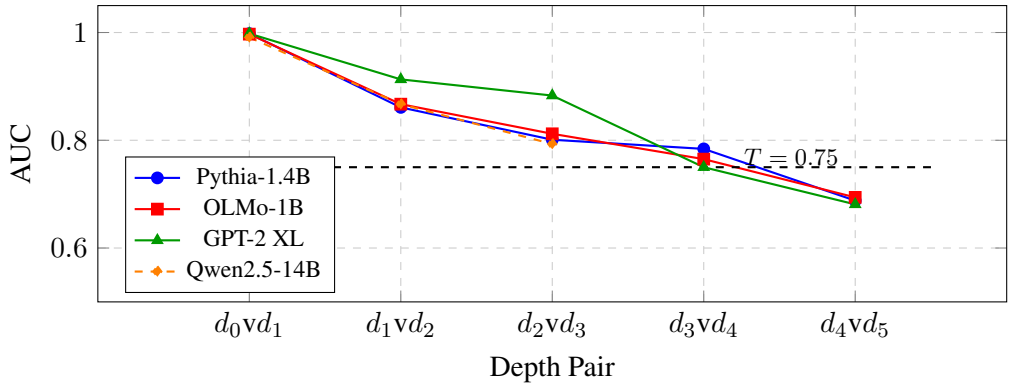


Figure 1: AUC decay across adjacent depth pairs for nucleus sampling. The dashed line marks the EDD threshold $T = 0.75$. AUC decays monotonically for all models. The 14B model (Qwen2.5, dashed orange) follows the same decay pattern as 1B models, confirming scale invariance. Beyond d_4vd_5, all models fall below threshold.

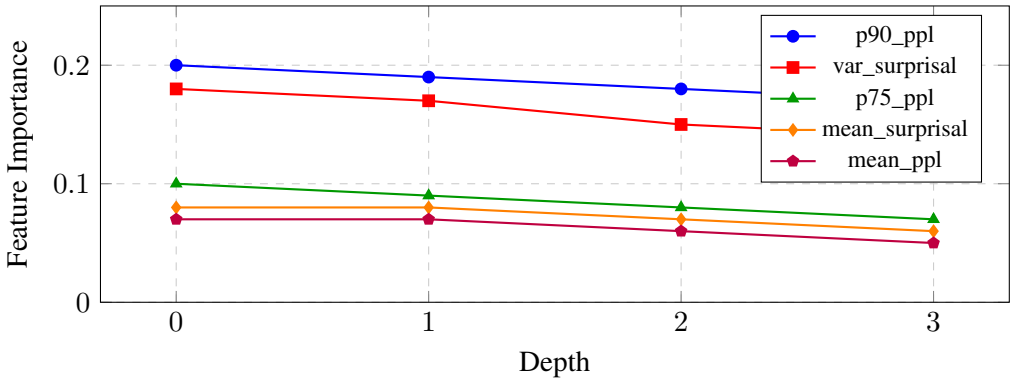


Figure 2: Top-5 feature importance across depths (GPT-2 XL, nucleus, RF). `p90_ppl` and `var_surprisal` consistently dominate regardless of which depth boundary is being classified. Importance decays slightly at deeper boundaries, consistent with reduced signal-to-noise as distributions converge.

the two-stage pipeline provides a practical mitigation. Cross-strategy deployment without prior strategy knowledge requires either the two-stage pipeline or strategy-invariant feature engineering, which remains an open challenge.

Reference Model Circularity. Our feature extraction uses Pythia-1.4B as the fixed reference model. When classifying text generated by Pythia-1.4B itself, there is a potential circularity concern: the reference model is the same as (an ancestor of) the generating model. Two observations mitigate this concern. First, cross-model LOO experiments (training on two models, testing on the third) achieve 62–69% accuracy, well above chance, demonstrating that depth features generalize across model families. Second, the 14B experiment shows that Pythia-1.4B retains discriminative power on text generated by a much larger model (Qwen2.5-14B), with only a 2–4 percentage point gap versus within-scale classification.

Temporal Signal in Web Data. The 17.4 percentage point increase in d_1+ classification for post-2023 web text (Section 5.5) is consistent with the timeline of widespread LLM deployment but should be interpreted cautiously. Changes in web content composition (more template-based and SEO-optimized content independent of LLMs) could contribute to this shift. The cross-model agreement of 87–91% on wild data provides some confidence that the signal is real rather than classifier-specific, but ground-truth validation on labeled web corpora would strengthen the claim.

	$\hat{d}_0$	$\hat{d}_1$	$\hat{d}_2$	$\hat{d}_3$
d_0	**.985**	.010	.003	.002
d_1	.050	**.810**	.090	.050
d_2	.020	.150	**.580**	.250
d_3	.010	.060	.210	**.720**

Figure 3: Confusion matrix for 4-class depth estimation on Qwen2.5-14B (nucleus, RF). Depth 0 is near-perfectly identified (recall = 0.985). The primary confusion occurs between adjacent depths (d_1/d_2 and d_2/d_3), consistent with the monotonic decay in inter-depth discriminability. Depth 2 has the lowest recall (0.58), squeezed between neighboring depth distributions.

Feature	**Nucleus**	**Temp 0.9**	**Top-k 50**
`p90_ppl`	↓↓↓	↓↓↓↓	↓↓
`var_surprisal`	↓↓↓	↓↓↓↓	↓↓
`mean_ppl`	↓↓	↑ **rev**	↓
`TTR`	↓	↓↓	↓
`bigram_ent`	↓	↑ **rev**	↓

Figure 4: Atlas heatmap: feature trajectory direction and magnitude across strategies (1B models, aggregated). Arrows indicate the direction of feature change from $d_1 \rightarrow d_2$. ↓: features decrease with depth (expected). "rev" cells: temperature reverses the $d_1 \rightarrow d_2$ trajectory for mean perplexity and bigram entropy, the root cause of cross-strategy transfer failure.

8 Limitations

Cross-Strategy Deployment Constraint. When the decoding strategy is unknown, naive transfer of depth classifiers can produce reversed predictions at the d_1/d_2 boundary. The two-stage pipeline mitigates this (Section 5.4) but requires an initial strategy identification step. Deployment without strategy knowledge and without the two-stage pipeline is unreliable for intermediate depths.

Controlled vs In-the-Wild Gap. Our benchmark constructs clean recursive chains with a single model and strategy per chain. Real-world recursive contamination may involve heterogeneous models, strategies, post-editing, and partial human revision. The prompted generation experiment (Section 5.3) bridges toward ecological validity but does not fully close this gap.

Reference Model Dependency. All features are computed relative to Pythia-1.4B. While cross-scale results are encouraging, the optimal reference model for arbitrary target text is unknown, and reference model selection may affect performance on text from very different domains or languages.

Effective Discriminable Depth. EDD at $T = 0.75$ reaches 3–4 for most configurations, meaning depths beyond d_3 or d_4 are not reliably distinguishable. For practical curation, the distinction between shallow ($d \leq 1$) and deep ($d \geq 2$) recursive text may be sufficient, but fine-grained depth estimation at large depths remains infeasible with current features.

False Positive Rate on Web Data. Wild-data calibration shows 15–20% of web-crawl passages are classified as d_1+. Template content, SEO text, and other repetitive human-written patterns produce perplexity signatures that overlap with recursive synthetic text. Ensemble voting across reference models reduces but does not eliminate this issue.

Table 8: Cross-strategy LOO d_1vd_2 AUC. Values below 0.5 indicate decision boundary reversal. Z-score normalization does not recover the boundary.

Holdout Strategy	**GPT-2 XL**	**OLMo**
Temperature (raw)	0.13	0.25
Temperature (z-score)	0.24	0.19
Nucleus	0.82	0.79
Top-k	0.71	0.68

9 Conclusion

We formalized ordinal generation depth estimation as a new task for characterizing recursive synthetic contamination in text corpora. Our Depth Degradation Atlas provides the first systematic empirical map of how text statistical properties evolve across recursive depths, model scales from 1B to 14B parameters, three decoding strategies, and both unconditional and prompted generation modes. Feature-based classifiers achieve 70–85% four-class accuracy across all tested configurations, with perplexity tail statistics (`p90_ppl`, `var_surprisal`) serving as universal depth markers. The Atlas reveals that temperature sampling induces a structural feature trajectory reversal at the d_1/d_2 boundary, a finding with direct implications for cross-strategy deployment. Our two-stage pipeline addresses this challenge, achieving 75.2% accuracy by routing samples through strategy identification before depth classification. Wild-data calibration on web corpora shows consistent and interpretable depth distributions that align with known quality hierarchies and temporal trends in LLM-generated content. Together, these contributions establish ordinal depth estimation as a viable and informative tool for training data curation in the era of recursive synthetic content.

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